

Steps of SMOTE (Synthetic Minority Over-sampling Technique)

Step-by-Step Explanation of the Flow Chart

1. **Load the dataset** containing the input features and target labels.
2. **Check the class distribution** to determine whether the dataset is balanced or imbalanced.
3. If the dataset is already balanced, proceed directly to **model training**.
4. If the dataset is imbalanced:
 - Identify all **minority-class samples**.
 - For each minority sample, find its **k nearest minority neighbors**.
 - Randomly select one of these neighbors.
 - Generate a random value λ between **0 and 1**.
 - Create a new synthetic sample using the SMOTE formula:

$$\text{New Sample} = A + \lambda(B - A)$$

where:

- A = Current minority sample
 - B = Selected nearest minority neighbor
5. Add the newly created synthetic sample to the dataset.
 6. Repeat the process until the desired number of minority samples is generated.
 7. Train the machine learning model on the **balanced dataset**.
 8. Evaluate the model using suitable performance metrics such as **Accuracy, Precision, Recall, F1-score, and Confusion Matrix**.

```
[22]: from imblearn.over_sampling import SMOTE
import pandas as pd

# Original dataset
data = {
    'Study_Hours': [1, 2, 3, 4, 5, 6, 7],
    'Result': [0, 0, 0, 0, 0, 1, 1]
}

df = pd.DataFrame(data)

X = df[['Study_Hours']]
y = df['Result']

print("Original Class Distribution")
print(y.value_counts())

# Apply SMOTE
smote = SMOTE(random_state=42, k_neighbors=1)
X_resampled, y_resampled = smote.fit_resample(X, y)

print("\nAfter Applying SMOTE")
print(y_resampled.value_counts())

# Display the new dataset
resampled_df = pd.DataFrame(X_resampled, columns=['Study_Hours'])
resampled_df['Result'] = y_resampled

print("\nResampled Dataset")
print(resampled_df)
```